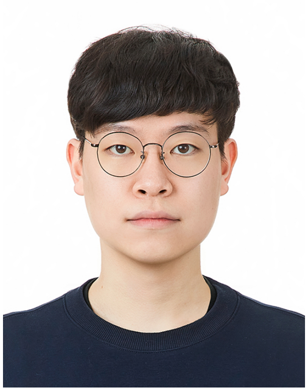
Jonghyun Woo

Email · GitHub · LinkedIn · Youtube



Research Interests

- **Control theories** with **optimization** and **estimation**
- **Applications of AI** in control dynamic systems

Education

- **M.S. in Mechanical Engineering** (Mar 2025 - Present)
  Chung-Ang University (CAU)
  Major: Robotics and Control Engineering
  Advisor: Prof. Seokwon Lee, Autonomous and Intelligent Systems Lab (AISL)
  GPA: In progress / 4.5
  Full Scholarship (Graduate Research Scholarship), CAU

- **B.S. in Mechanical Engineering** (Mar 2019 - Feb 2025)
  CAU
  GPA: 4.16 / 4.5 (*Graduated cum laude*)
  Full Scholarship (Davinch Scholarship), CAU

Experience

- **Teaching Assistant** (Mar 2025 - Present)
  CAU

- **Undergraduate Researcher** (Jun 2023 - Feb 2025)
  AISL, CAU

- **Embedded Software Developer** (Mar 2019 - Feb 2025)
  MACH (Aerospace Engineering Club), CAU

- **Vehicle Mechanic** (Dec 2020 - May 2022)
  The Republic of Korea Army

Publication

International Journal

1. Jonghyun Woo, Inyoung Jung, and Seokwon Lee, "A Comprehensive Framework for Modelling and Control of Morphing Quadrotor Drones," Aerospace, Under review.
2. Jonghyun Woo, Jongho Park, and Seokwon Lee, "Reactive Collision Avoidance for Small Unmanned Aerial Robots Using Probabilistic Ellipsoidal Geometry and Bi-Directional Collision Cone" Nonlinear Dynamics, 2025, under review.

International Conference

1. Jonghyun Woo, and Seokwon Lee, "Integrated Probabilistic and Geometric approaches for Reactive Collision Avoidance in Unmanned Aerial Vehicles ", 2025 Europe-Korea Conference on Science and Technology (EKC 2025), Vienna, Austria, August 2025.

Domestic Conference

1. 김영호, 이석원, 김형빈, 우종현, “안전 영역 및 입력 포화를 고려한 계층적 제어장벽 함수 기반 인공위성의 랑데뷰 유도기법 설계”, 한국항공우주학회 우주학술대회, 디오션 여수, 여수, 전라남도, 2025년 6월.
2. 우종현, 이석원, “동적 장애물에 대한 멀티콥터의 타원 기하 기반 충돌 회피 알고리즘”, 한국항공우주학회 추계학술대회, 하이원리조트 컨벤션호텔, 정선군, 강원도, 2024년 11월.

Projects

- **Robust Attitude Controller of Tilt-rotor VTOL AIRBILITY** (Jul 2025 - Dec 2025)
  Developed a robust controller for tilt-rotor VTOL using ADRC.
- **Trajectory Optimization of Reusable Unmanned Space Vehicle (ReUSV)** (Feb 2025 - Dec 2025)
- **Magnetometer Calibration of Wearable Robot HUROTICS** (Jun 2025 - Aug 2025)
  Implemented VQF algorithm based magnetometer calibration for AHRS of wearable robot.
- **Morphing Drone** (Github, Youtube) (Dec 2024 - Nov 2025)
  Developed a simulation of a drone capable of morphing its structure using LQR control.
- **Wheeled Bipedal Robot (WBR)** (Github, Youtube) (Sep 2024 - Jan 2025)
  Designed a WBR inspired by Ascento.
- **7-DoF Manipulator** (Github, Youtube) (Sep 2024 - Dec 2024)
  Implemented motion planning and control algorithms for Panda using ROS.
- **VTOL** (Github) (Dec 2023 - Jul 2024)
  Designed and simulated a vector-field guidance control to VTOL.
- **Balancing Robot** (Youtube) (Mar 2023 - Jun 2023)
  Designed and implemented a balancing robot using PID control.
- **PID Control Implementation** (Youtube) (Sep 2022 - Dec 2022)
  Implemented PID control on seesaw system, beam and ball system, and tower-copter system.

Awards

- Mechanical Engineering Short Term Research (MESTeR), CAU, Mentor, 1st Place (Mar 2025)
- Competition for Capstone Design, CAU, 1st Place (Dec 2024)
- MESTeR, CAU, Researcher, 1st Place (Jun 2023)
- ALCP Program, CAU, 2nd Place (Dec 2022)
- Design and Flight of UAV Competition, PNU, 3rd Place (Dec 2020)
- Design and Flight of UAV Competition, GNU, 2nd Place (Aug 2019)

Skills

- **Programming Languages**
  - MATLAB
  - Python
  - C/C++
- **Engineering Tools**
  - CAD (*Inventor, CATIA*)
  - 3D-Printing (*3DWOX, Bambu-lab, Anycubic*)
  - VICON
- **Other Skills**
  - Video editting (*iMovie, LumaFusion, ClipChamp*)
  - Multicopter Pilot